

CSE-355 (Spring 2026)

HW 01

Homework 1

Total Marks: 100

Deadline: Feb 22, 2026 at 11:55pm

General Instructions

- There are a total of 10 questions and each question carries 10 marks. (some are easier than the others and this has been done on purpose)
- This assignment must be completed and submitted individually. Group submissions are NOT allowed
- Simply dividing questions among peers or sharing written solutions is strictly prohibited and defeats the purpose of the assignment
- Every student is expected to develop a complete and independent understanding of all questions
- You may use online resources, including Generative AI tools such as ChatGPT, strictly for learning and clarification purposes. However, copying solutions, generating answers without understanding them, or submitting AI-produced content verbatim is considered academic misconduct
- If you have any confusion, please post your questions on the Whatsapp Group of the course or contact the TA or the Instructor directly

Submission Instructions

- All solutions must be typed and submitted in PDF format. **Handwritten submissions will NOT be accepted under any circumstances.**
- Typesetting your solutions in **LaTeX** is strongly encouraged and preferred. However, you can also use MS Word or Google Docs, etc.
- Submissions must be made on LMS before the stated deadline. Email submissions will NOT be accepted. The deadline will not be extended without imposing a penalty for late submission.
- Each student must submit their own assignment individually. The first page must clearly include your full name and student ID.

Question 1

Suppose Alice wants to send a confidential message to Bob. She does not want anyone else to see what she sent. Alice also wants Bob to be sure it really came from her.

How can we meet all the requirements? Explain by specifying all the actions taken by both Alice and Bob. Feel free to simply draw a block diagram.

Solution:

- 1) Alice: Message $\rightarrow$ Hash $\rightarrow$ Alice Private Key $\rightarrow$ Signature
 - 2) Alice: Message $\rightarrow$ Bob Public Key $\rightarrow$ Encrypted Message
 - 3) Send output of (1) and (2): Signature and Encrypted Message
 - 4) Bob: Encrypted Message from (2) $\rightarrow$ Bob Private Key $\rightarrow$ Hash $\rightarrow$ Output
 - 5) Bob: Signature from (1) $\rightarrow$ Alice Public Key $\rightarrow$ Output
- Bob compares the outputs of (4) and (5). If they are the same, every requirement met!

Question 2

Bob receives a message that claims to be signed by Alice. He tries to verify it using Alice's public key, but the verification fails.

a) What does this outcome mean?

Solution: The verification failure means that either the message was not signed by Alice's private key or the message was altered after it was signed. The signature does not match the message content when verified with Alice's public key.

b) Based on this, what possible issues or attacks should Bob suspect?

Solution: Bob should suspect either: 1) The message is a forgery (not actually from Alice), 2) The message was tampered with by a man-in-the-middle, or 3) There was a transmission error that corrupted the message or signature.

Question 3

a) Alice comes up with her own digital signature scheme. She writes a message and encrypts that message using her Private key. She then sends this message and the encrypted message both to Bob. Can Bob be sure that this message came from Alice?

Solution: No. As mentioned in class, Kevin may use Alice's *public* key to generate garbage data and send it to Bob.

b) Alice modifies this scheme. She writes a message and encrypts that with her private key (as in the previous part). She then hashes the encrypted message and sends this hash along with the original message to Bob. Can Bob be sure that this message is from Alice?

Solution: No. Bob has no way of verifying whether the message was, indeed, originally from Alice or otherwise. This is because the hash function is only one way – and Bob can never know the encrypted intermediate message (since he doesn't have Alice's private key).

Question 4

In class we learnt that a symmetric key is often used for communication *in-bulk* i.e., when there is a lot of data that must be transferred. This is done because symmetric keys operate a lot faster than asymmetric keys.

a) Assume that Alice and Bob share a secret, symmetric key. Devise a scheme with which Alice can send a plain-text message to Bob and ensure that Bob can verify the authenticity of this message.

Assume that Alice and Bob must share a LOT of data and asymmetric keys are not viable.

Solution: The scheme we are trying to invent is very similar to that of Message Authentication codes. Since Bob and Alice share a single symmetric key and privacy is not a concern, only authenticity, we just need to make sure our message has not been tampered with.

Alice generates message M and takes its hash $h(M)$. Then she “signs” it by applying the symmetric key to create a signature S . She sends the M and S to Bob.

To verify, Bob 1) takes the hash of message M and, 2) applies the symmetric key to the signature S . If the result of these two operations match, the message is from Alice.

b) Consider a scenario in which Alice wishes to relay to Bob the message “Give me 3.50\$”. A malicious actor, Kevin, duplicates this message and sends it twice to Bob (this is called a replay attack). According to the scheme you devised in the previous part, what will the outcome be? Further, devise a scheme in which duplicated messages are rejected.

Solution: Kevin can duplicate Alice’s messages and Bob has no way of knowing.

However, there is an easy fix to this, add a nonce or a timestamp to the messages. If a message shows up with the same nonce or timestamp, Bob should reject it.

Question 5

In a Bitcoin blockchain, a transaction claims “Alice pays 0.3 BTC to Bob.”

a) What checks must be performed on Alice’s digital signature before this transaction is considered valid?

Solution: Before the transaction is considered valid, the following checks must be performed on Alice’s digital signature: 1) Verify that the signature is mathematically valid using Alice’s public key, 2) Confirm that Alice had sufficient funds to make this transaction, 3) Ensure this transaction hasn’t already been spent (avoiding double-spending).

b) Could Alice later deny sending it? Why or why not?

Solution: No, Alice cannot plausibly deny sending it later due to the non-repudiation property of digital signatures. The valid signature serves as cryptographic proof that only someone with access to Alice’s private key could have created that signature.

Question 6

Consider a blockchain with three blocks. Block 1 contains transactions and has a hash $H1 = \text{ABC123}$. Block 2 contains its own transactions and includes $H1$ in its data before being hashed to produce $H2 = \text{DEF456}$. Block 3 does the same, resulting in $H3 = \text{GHI789}$.

a) If a transaction in Block 1 is altered, explain why merely recalculating its hash to $H1'$ is insufficient to hide the tampering.

Solution: Merely recalculating Block 1's hash to $H1'$ is insufficient because Block 2 contains $H1$ in its data. If $H1$ changes to $H1'$, then the hash of Block 2 must also be recalculated since it includes the previous block's hash as part of its input data. This would change $H2$ to something different, which would then require recalculating $H3$, and so on for all subsequent blocks.

b) What must an attacker do to make the altered chain appear valid to a node synchronizing from the latest block? Why does Proof-of-Work make this practically very difficult?

Solution: To make the altered chain appear valid, an attacker must not only recalculate all hashes from the altered block forward but also redo the Proof-of-Work for each block. Proof-of-Work makes this practically difficult because it requires substantial computational effort (mining) for each block. The attacker would need to outperform the entire honest network's computational power to create a longer valid chain, which is economically infeasible for established blockchains.

Question 7

Consider a simple encryption procedure using a public/private key pair. You have the public key as the integer $p=7$ and the "secret" private key as the integer $s=3$.

The method to encrypt an integer m is: $c = m^p \bmod 33$

And the method to decrypt an encrypted integer is: $m = c^s \bmod 33$

Assume that the input characters have the following numerical values:

$a/A = 0, b/B = 1, c/C = 2, \dots, z/Z = 25$.

a) Encrypt the word 'SurpriseQuiz' and write down the sequence of numbers.

Solution: If we convert it to numerical values, we

get: 18 20 17 15 17 8 18 4 16 20 8 25.

If we apply the public key to this, we get:

6 26 8 27 8 2 6 16 25 26 2 31

b) Decrypt the sequence "1 2 13 29 20 2 7 20 21 16 8 16 13 28". Write down the characters and what you think of them.

Solution:

This translates to "BitcoinOverEth"

Now let's consider a hashing function $h(x) = \lfloor e^x + 100 \rfloor \bmod 32$ where $x \in [0, 50]$.

As $h(x)$ is defined only for one input, we will modify it to take in lists of numbers such as "31 23 4 2 13". We define that $h(31\ 23\ 4\ 2\ 13)$ will be $h((31 + 23 + 4 + 2 + 13) \bmod 50)$. The idea is that every number in the list makes a contribution. Suppose Alice has the public key $p = 7$ and private key $s = 3$. Data is sent in the form " $m_1 m_2 m_3 \dots m_n \text{ sig}$ " where the m_i are numbers from the message and "sig" is the **private** key (encryption) of the hash of the

message. Note that this is different from the previous parts.

c) Suppose Alice wants to transmit the word “SaveMe”. What is the final message that Alice transmits?

Solution: The message is “18 0 21 4 12 4”. We compute the hash as

$$h((18+0+21+4+12+4) \bmod 50) = h(59 \bmod 50) = h(9).$$

$$\text{Then } h(9) = \lfloor e^9 + 100 \rfloor \bmod 32$$

this gives $h(9) = 11$. Then the private key encryption on 11 is $11^3 \bmod 33 = 11$.

So the final message is “18 0 21 4 12 4 11”.

d) You receive a message “12 20 18 19 0 5 0 17 14 2 10 18 1 0 2 7 4 18 7 14 2 10 18 5”. It is known that the scheme in part (c) is used. Is this message from Alice? Why or why not? If it is a message from Alice then write the message.

Solution:

The hash of the message is: $12+20+18+19+0+5+0+17+14+2+10+18+1+0+2+7+4+18+7+14+2+10+18 = 218$ $h(218 \bmod 50) = h(18) = 5$. Now the public key (decryption) on the signature is 14 and it does not match the hash of the message, so this message is not from Alice. message: “MustafaRocksBacheShocks”

Question 8

a) What is the fundamental difference between Symmetric and Asymmetric (Public Key) cryptography?

Solution: The fundamental difference is that symmetric cryptography uses the same key for encryption and decryption, while asymmetric cryptography uses a pair of mathematically related keys: a public key for encryption and a private key for decryption (or a private key for signing and public key for verification).

b) For the following goals, state whether symmetric or asymmetric cryptography is typically used. Also state your reasoning: - i) Encrypting a large file for storage. - ii) Proving your identity to a website. - iii) Establishing a secure connection to your bank’s server.

Solution: i) Symmetric cryptography - More efficient for encrypting large amounts of data.
 ii) Asymmetric cryptography - Allows authentication without sharing secret keys.
 iii) Both - Typically uses asymmetric crypto to establish a shared secret (symmetric key) which is then used for the bulk encryption of the session (combining benefits of both approaches).

Question 9

A simple hash function is defined as $H(x) = (5x + 3) \bmod 16$.

a) Calculate the hash of the following numbers: $H(7)$, $H(23)$, $H(39)$.

$$\text{Solution: } H(7) = (5 \times 7 + 3) \bmod 16 = (35 + 3) \bmod 16 = 38 \bmod 16 = 6$$

$$H(23) = (5 \times 23 + 3) \bmod 16 = (115 + 3) \bmod 16 = 118 \bmod 16 = 6$$

$$H(39) = (5 \times 39 + 3) \bmod 16 = (195 + 3) \bmod 16 = 198 \bmod 16 = 6$$

b) What do you observe about the results for $H(7)$ and $H(23)$? Explain why this happens, relating it to the properties of the modulo operation.

Solution: All three inputs (7, 23, 39) produce the same hash output (6). This happens because 7, 23, and 39 are all congruent modulo 16 ($7 \bmod 16 = 7$, $23 \bmod 16 = 7$, $39 \bmod 16 = 7$), and the hash function is linear. This demonstrates a lack of avalanche effect and poor collision resistance.

c) Based on this, what is a major weakness of this hash function?

Solution: A major weakness is its extremely poor collision resistance - many different inputs produce the same output. This makes it vulnerable to collision attacks and unsuitable for cryptographic purposes.

Question 10

We briefly talked about Proof-of-Stake (PoS) as an alternative to Proof-of-Work (PoW).

a) In PoW, miners expend energy. What do validators in a PoS system “stake” or put at risk to participate in block creation?

Solution: In Proof-of-Stake systems, validators “stake” their own cryptocurrency tokens as collateral. This stake is put at risk - if they validate fraudulent transactions, they can lose part or all of their staked tokens through a process called “slashing.”

b) Read up on the advantages and criticisms of PoS systems, and mention one for each here.

Solution: PoS is much more energy efficient than PoW since it doesn't require massive computational power. PoS may lead to wealth concentration where the richest stakeholders have more influence over the network, potentially leading to centralization.